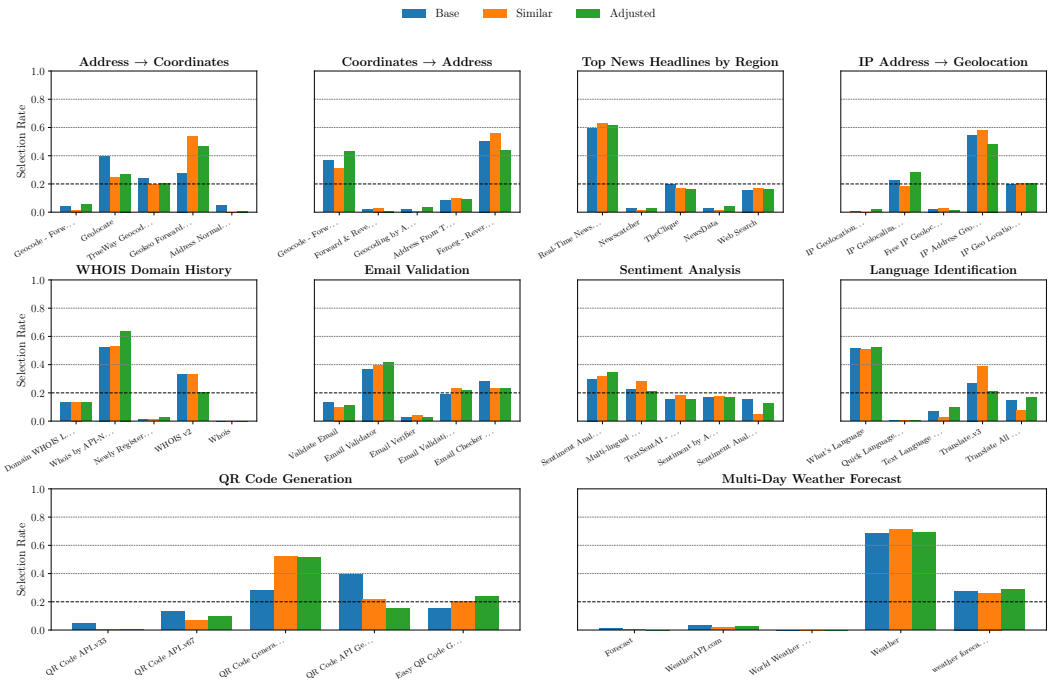

Figure 12: Selection distributions for ChatGPT 3.5 using different system prompts across ten clusters. Each subplot corresponds to one cluster, with the x-axis indicating the API in the cluster and the y-axis showing the fraction of times that API was chosen using the corresponding system prompt over 500 runs.

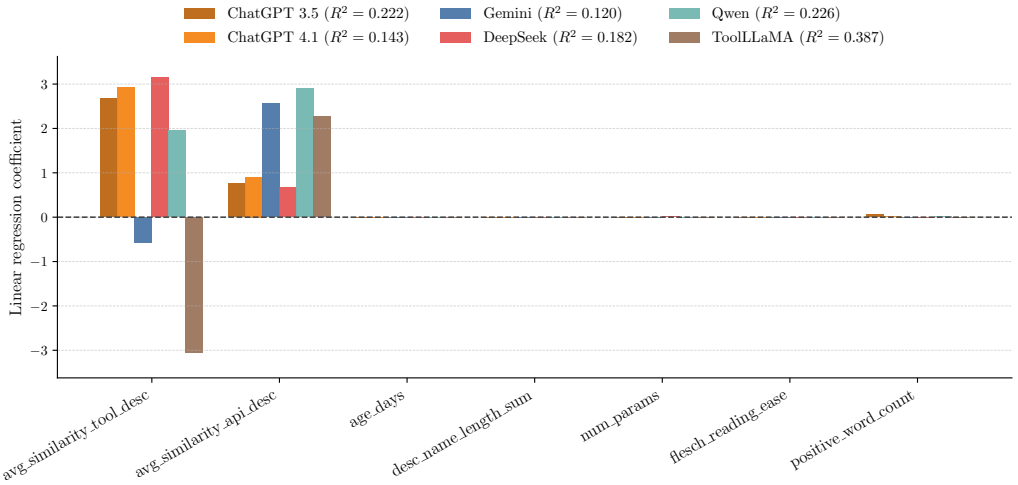

Figure 13: Linear regression feature weights used to predict API selection rates for six LLMs. Each group of bars corresponds to one API-level feature; different colors denote models, with their R^2 shown in the legend. Larger positive weights indicate features that increase the predicted selection rate.

Random forests fail to offer meaningful improvement. We fitted random-forest regressors with the same mean-centered features using both cross-validation and a held-out split, but predictive performance was poor, meaning the forests often did worse than a trivial constant baseline. This suggests the available features, at least in their current form and scale, don't contain enough signal or that some artifacts are overwhelming the gains from nonlinearity. Therefore, any feature-importance estimates from these trees would be unreliable and we do not lean on them for explanation. Future work could involve revisiting this with a richer feature set, more data, or alternative nonlinear modeling.

F.2 How do metadata interventions affect API selection?

We saw that corrupting descriptions produces much larger and more stable effects on selection behavior than name-level perturbations, which are sometimes noisy and unpredictable, in Section 4.3. Figure 14 corroborates this across clusters. Name perturbations often leave distributions near-unchanged or can make them drift unpredictably (e.g., Cluster 1, where Cluster 1 is positioned at the top-left and Cluster 6 at the bottom-right), whereas description/parameter scrambles frequently overhaul rankings: sometimes amplifying the dominant API (Cluster 2), other times causing dramatic re-ordering (Clusters 3). Name edits rarely produce comparably stable re-ranking.

Together, these patterns indicate that description-level semantics (and, to a lesser extent, parameter semantics) are the primary cues models use to discriminate among functionally similar APIs. Name perturbation alone tend to inject noise without consistent effects. Finally, bias persists under minimal semantic signal (only names and schema fields), implying selection behavior sometimes relies on residual, non-obvious priors rather than solely on coherent, human-interpretable heuristics.

Manipulating the description of certain tools has mixed effects across clusters. Figure 14 (lower row) shows three behaviors when we manipulate descriptions. First, swapping the most- and least-popular tools' descriptions can invert their selection rates (Cluster 4), indicating description text alone can dominate choice. Second, the same swap sometimes yields only a modest lift for the least-popular tool while unexpectedly altering the selection shares of unaffected tools (Cluster 5), suggesting the landscape is reconfigured rather than ranks simply exchanged. Third, in some clusters the swap has minimal effect (Cluster 6), implying other cues—e.g., name priors or parameter schemas—anchor preferences.

Targeted corruption of the most-selected tool's description has similarly inconsistent effects. In Cluster 4, scrambling collapses its share to near zero as another tool absorbs the mass; in Clusters 5–6, corruption diffuses probability across competitors, producing a more even allocation. Overall, description tampering often wields substantial influence, but the impact is context-dependent: the same intervention can invert, redistribute, or barely change preferences, underscoring that tool choice emerges from multiple interacting cues.

G Additional Figures

G.1 Correlation in Selection between Models

Figure 15 shows that models exhibit varying degrees of correlation in their tool-selection patterns, suggesting shared but non-identical biases across families.

G.2 Selection Distributions for all Clusters

This subsection provides a full version of the figure referenced in the main text (Figure 3). It expands the subset plot to all ten clusters and keeps axes, run counts, and error-bar conventions identical to the summary in Section 4.2. Use Figure 16 for detailed inspection of per-cluster behavior.

G.3 Full Figure related to the CPT Experiment

Figure 17 shows how biased continued pre-training gradually increases preference for the exposed endpoint.

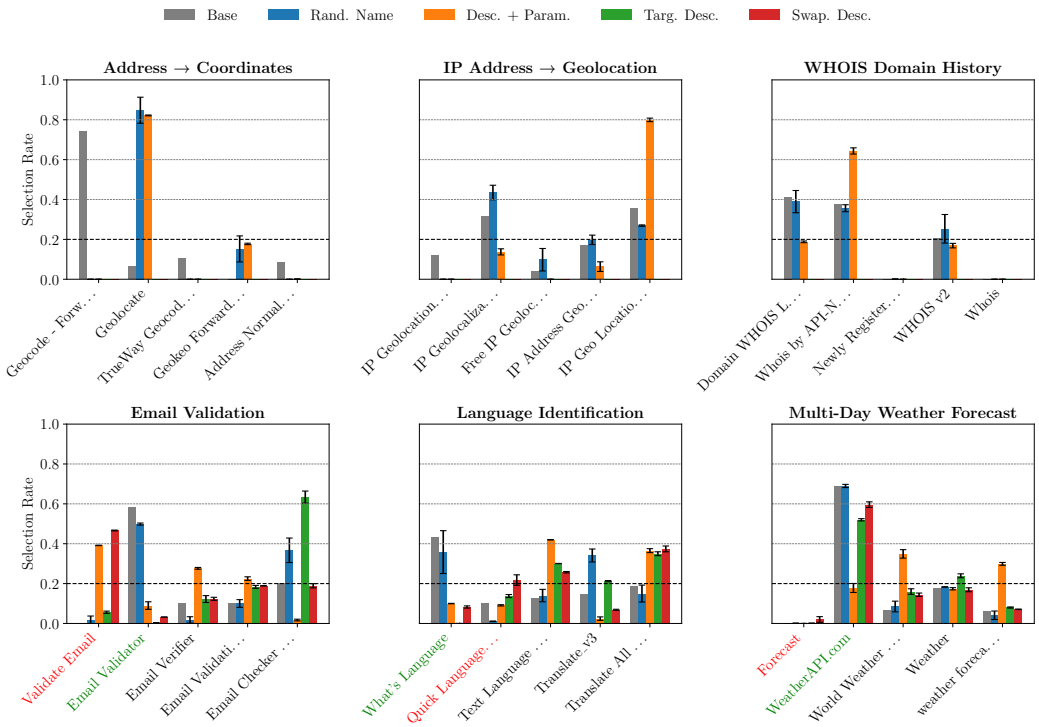

Figure 14: Selection distributions for Gemini under different name/ordering perturbations across six clusters of functionally equivalent APIs. Each subplot corresponds to one cluster; the x-axis lists the APIs and the y-axis shows the fraction of times the model under each condition selected that API, averaged over repeated runs. Error bars (when present) indicate the standard deviation across those repeats, making visible how robust or variable the preferences are under the different perturbations. Tools whose names are in green are the most selected by the baseline, and those in red are the least selected. These are the tools that are targeted for the swapping and selected scramble experiments.